AI HORIZON 2026 · CHALLENGE 2 · NSW AUTOMATION

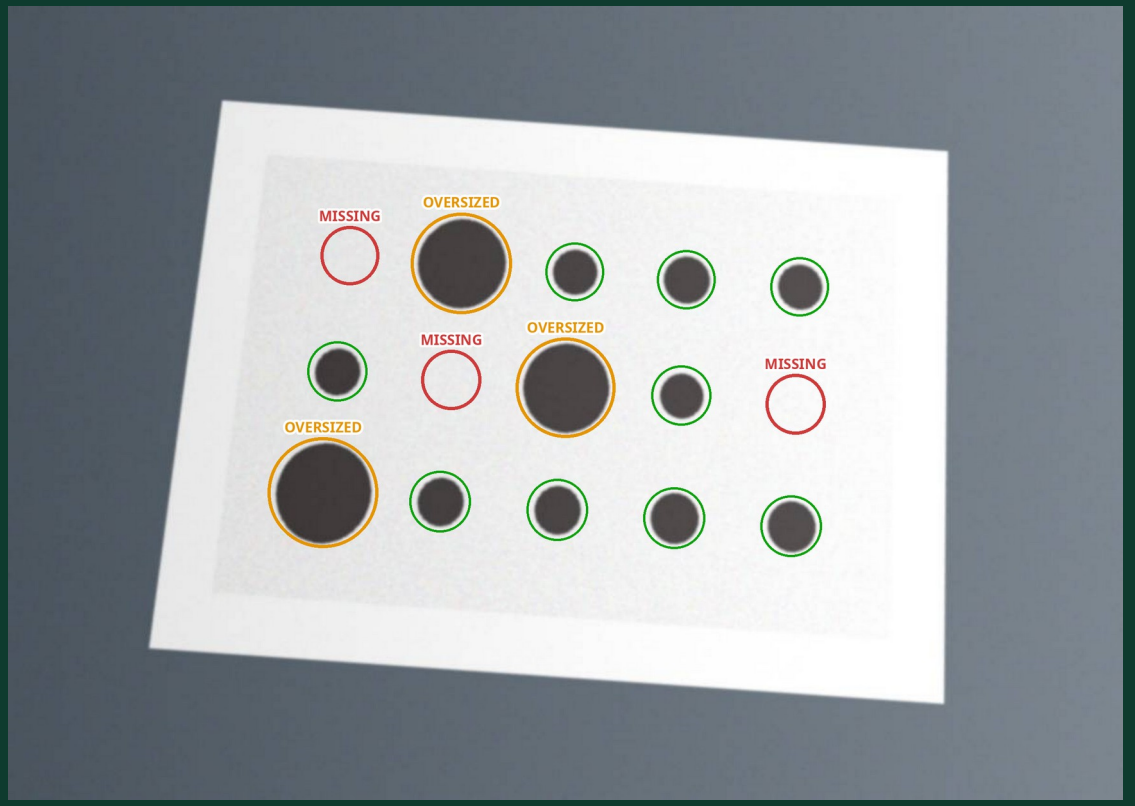
AI Dispensing Defect Detective

A phone photo and five questions → the likely cause, the reasons behind it, and what to check first.

And a case history that only learns from fixes that were proved.

All 5 steps · dynamic questions · all 4 bonus challenges

Team Lithium · Universiti Sains Malaysia



A phone photo of a printed sample card, lying on a desk. Each missing deposit is followed by an oversized one: the system reads carryover, p = 0.010.

THE PROBLEM

Dispensing defects are diagnosed by experience — and experience doesn't scale

New technicians guess

Inconsistent size alone has five plausible causes. The first check becomes a hunch — and a wrong strip-down costs a shift.

Three changes at once

Re-prime, raise the pressure, swap the needle. The defect goes away, and nobody can say which change did it.

Knowledge walks out

What the senior engineer knows is written down nowhere the next case can reach.

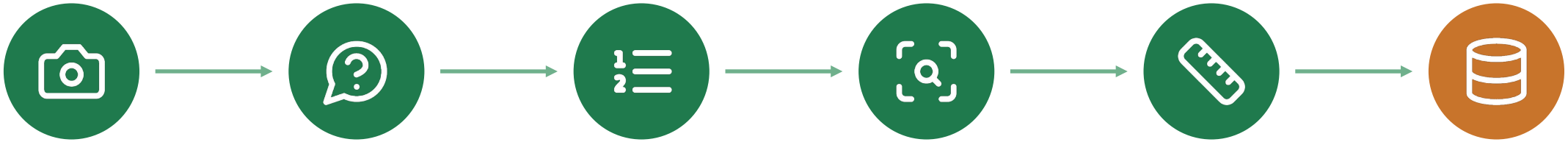
A score nobody can check

An engineer who can't see why a number says 85% won't act on it. Black-box confidence gets ignored on the line.

NSW Automation's brief: help engineers and technicians perform faster preliminary troubleshooting — not replace them.

THE SOLUTION

One guided session, from a phone photo to a proved fix



1 Photograph

Measures size spread, shape and missing dots — however the photo is framed

2 Interview

Five questions, then follow-ups picked because their answer changes the ranking

3 Rank

Seven causes from the manual — every % built from cited reasons

4 Check

Look first, then change one thing; the ranking updates live

5 Verify

Re-photograph: in spec on every check, held 20+ shots

6 Learn

Only proved fixes become precedent for the next case

WHAT IT ENABLES

- **A technician's first check, ranked with reasons**
- **An audit-ready PDF on every case**
- **Only proved fixes become precedent**
- **Offline on a line PC — a phone is enough**

THE AI

Two brains that are allowed to disagree — over a manual you can audit

Knowledge database

Seven causes and 42 weighted rules, every one citing NPL's Best Practice Guide by section and page. Past confirmed cases are counted as precedent: “similar problems occurred 12 times; in 8, the cause was trapped air.”

Dual-brain ranking

Pass 1 — a deterministic rule engine does the arithmetic in log-odds and sets every percentage. Pass 2 — an LLM critic reads the same evidence and can disagree. Disagreement lowers the tier and says so. The model never sets a number.

Senior-engineer loop

The operator escalates mid-case; the senior answers inside the case, logs the check that settles it, and can change a rule's weight — with a reason. It learns from people, not only from data.

96%

top cause correct when three or more reasons back it — 48% on one reason. The operator can count them on screen.

66% vs 14%

correct from one photo and five answers, with a quarter of the answers wrong — against a chance baseline of 14%.

108

automated checks: 79 backend tests, 27 measurement oracles, 2 end-to-end golden cases.

Precedent can lower confidence, never raise it — so the system cannot talk itself into being right.

RULE c.2 · THIRD-PARTY DECLARATION

What we built, and what we used

Our own work

- The fault tree and its weighted, cited rules
- Two-pass ranking, the chance and precedent gates, tiers T1–T4
- The five-condition close rule and the verdicts
- Five-axis interview and the follow-up selection
- Retrieval and precedent counting
- Photo registration, desk masking, all metrology
- The synthetic defect generator and the evaluation

Third-party, declared

- NPL Report MATC(A)82 — Best Practice for Electronic Materials Dispensing (2001), © Crown copyright, cited on screen
- NSW Automation challenge brief — rules citing “NSW brief p.N”
- Groq API, openai/gpt-oss-20b — optional critic, free-text mapping, written explanation
- FastAPI, Pydantic, SQLAlchemy, Alembic · OpenCV, NumPy
- Next.js, React, Tailwind, shadcn/ui, Base UI, lucide
- Dev only: pytest, Playwright, matplotlib

No model was trained or fine-tuned, and the vision uses no training data — it is classical OpenCV, reproducible by an engineer with a microscope.

LIMITATIONS

What it doesn't do — stated before you have to find it

No real production photos

NSW had none to give us, so every threshold is tuned on synthetic images. Validating on real NSW photos is step one of a pilot. This is our biggest gap.

Seven causes

Stringing, smearing and satellites (up height, NPL §3.3) have no cause yet. They're reported as “no cause ahead” and escalated — not guessed.

Busy boards

A trace touching a deposit can read as a distorted shape: 9% false alarms under light clutter. The fix is a bare-board reference photo.

Pixels, not microns

Everything is relative to the recipe; absolute volume needs a scale reference or a known pitch.

The LLM isn't reproducible

Critic and explanation can vary between runs — which is exactly why neither is allowed to change a likelihood.

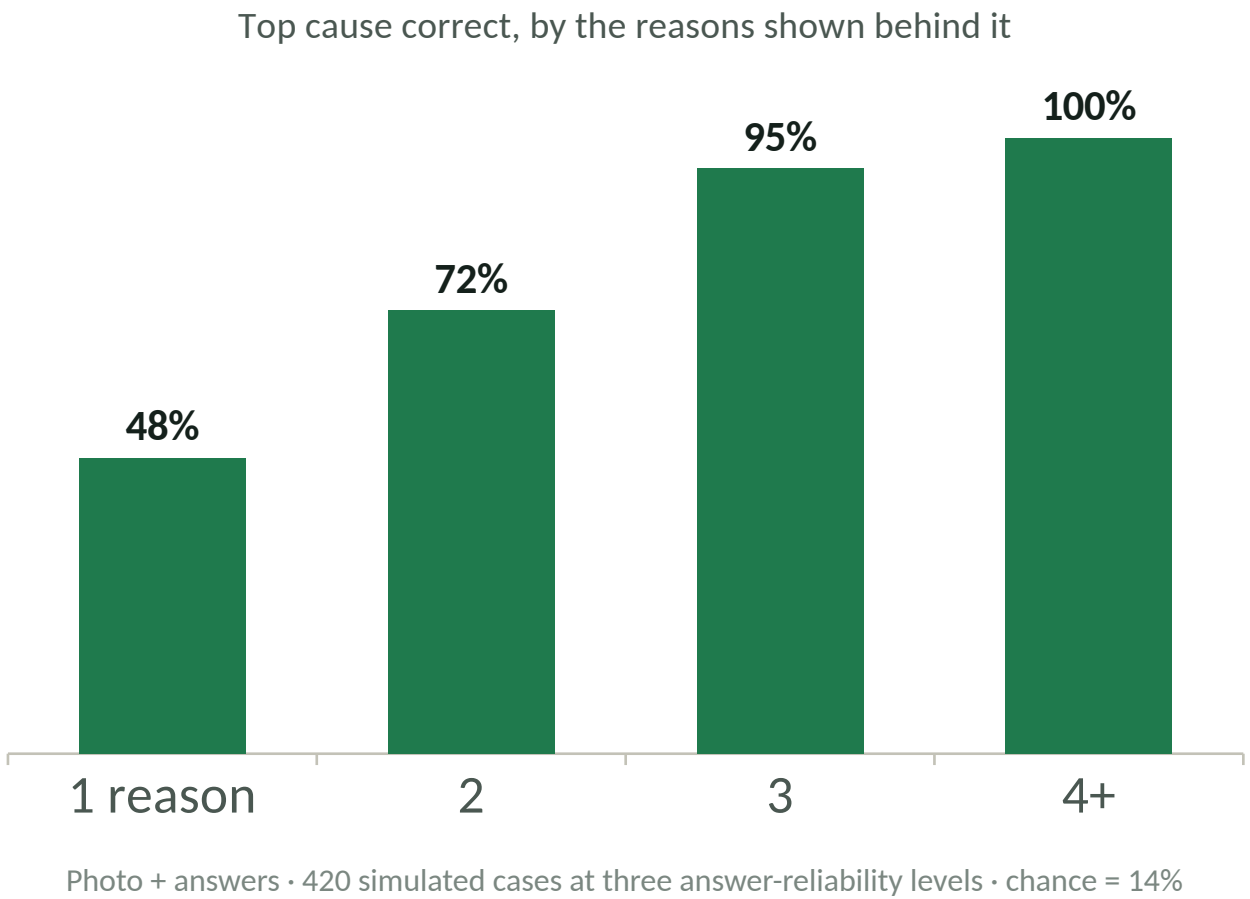
The measurement oracle deliberately includes cases the system fails, asserted as failures rather than hidden.

Faster first checks.
An auditable record.
A history that only learns from proved fixes.

Live demo →

BACKUP · EVIDENCE

The more reasons behind it, the more often it's right



66%

top cause right from one photo and five answers, with an operator whose answers are only 75% reliable — chance is 14%

6,600

synthetic photos and 420 simulated cases through the real code, in every evaluation run (about a minute)

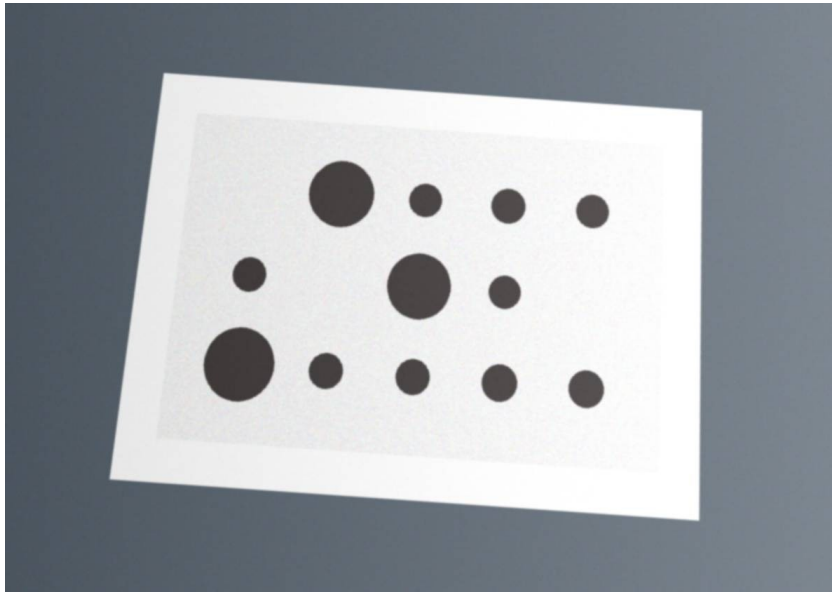
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automated checks — 79 backend tests, 27 measurement oracles, 2 golden cases — plus a browser run of the whole wizard

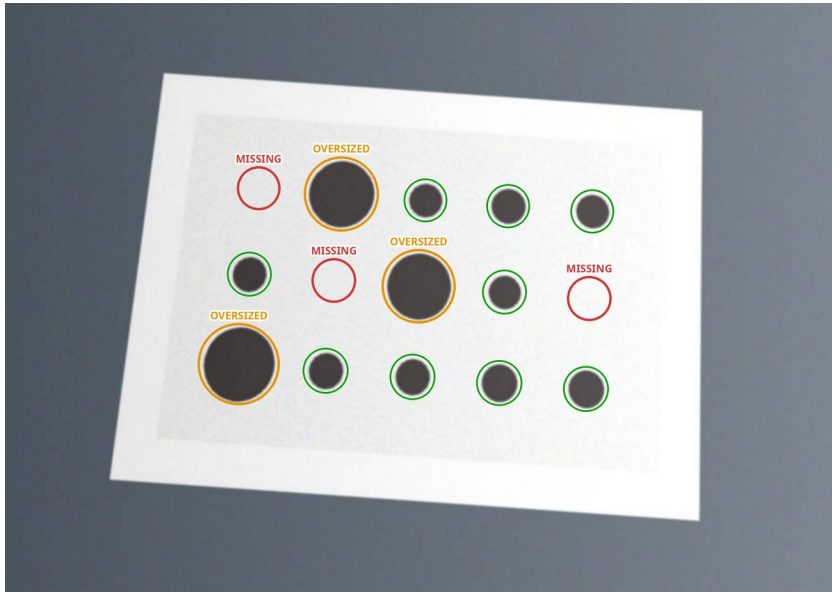
All test images are synthetic — NSW provided no production photos. Method, limits and every table: evaluation/README.md.

BACKUP · FEASIBILITY

Works on the photo an operator actually takes



Phone photo: card on a desk, camera tilted and off-centre



Recipe lined up (scale, rotation, perspective), then measured: 3 missing, 3 oversized, carryover p = 0.010

0% → 89%

of phone-style photos measured correctly, before → after registration — the same as perfectly framed ones

0

false alarms from zoom, framing, tilt or a desk in shot

Photo simulated from a printed sample card (the evaluation's 'on desk' condition). Validating on real NSW photos is step one of the pilot.

BACKUP · TRUST

Closing a case is not the same as fixing it

CONFIRMED

All five proved

- a check confirmed one cause
- exactly one change, aimed at it
- re-measured against a 'before' photo
- in spec on every check
- held 20+ shots

The only cases future rankings learn from

PLAUSIBLE

Some proved

- e.g. cause certain (tier T1) — but one deposit is still missing after the fix

Shown as context; adjusts nothing

UNVERIFIED

Nothing proved

- filed with the gaps named — e.g. '3 things changed at once: no way to say which one worked'

Adjusts nothing

T1 says how sure the ranking is. CONFIRMED says the fix was proved.

BACKUP · CONFIDENCE TIERS

Four confidence tiers — and agreement never raises one

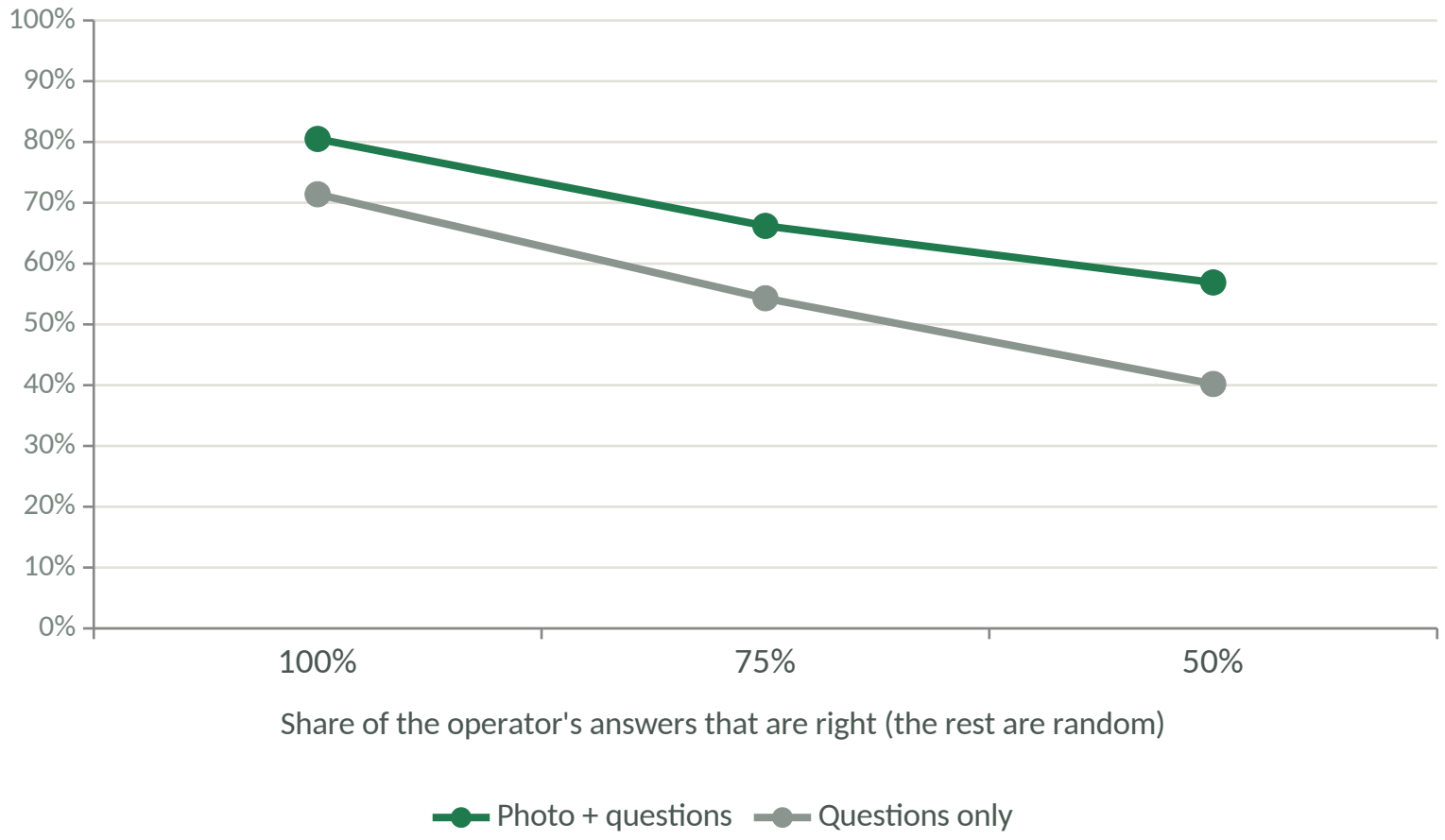
First: did a check on the machine confirm the top cause? Then it's T1, whatever the brains or gates say. → T1

	Measured pattern beats chance Gate A · small-then-big pattern, p < 0.05	Nothing measured decides a size spread alone is shared by several causes
Brain 2: no objection agrees, isn't sure, or has no LLM key	**T1 Strong** Plus 3+ matching confirmed cases, none disagreeing (Gate B). Otherwise T2. **Shows:** one lead cause, with its evidence	**T2 Provisional** No measurement singles the lead out. **Shows:** the lead, marked provisional, and the check to confirm it
Brain 2 objects a clear, different top pick	**T3 Disagreement** The measurement still backs Brain 1's cause. **Shows:** both candidates, Brain 1's on top	**T4 Don't force it** Nothing measured separates the two causes. **Shows:** no lead cause, just the check that settles it

Every tier says why on screen: *"Provisional: fewer than 3 matching confirmed cases back it. A check that confirms it would settle it."*

BACKUP · EVALUATION

The photo matters most when the answers are worst



57% vs 40%

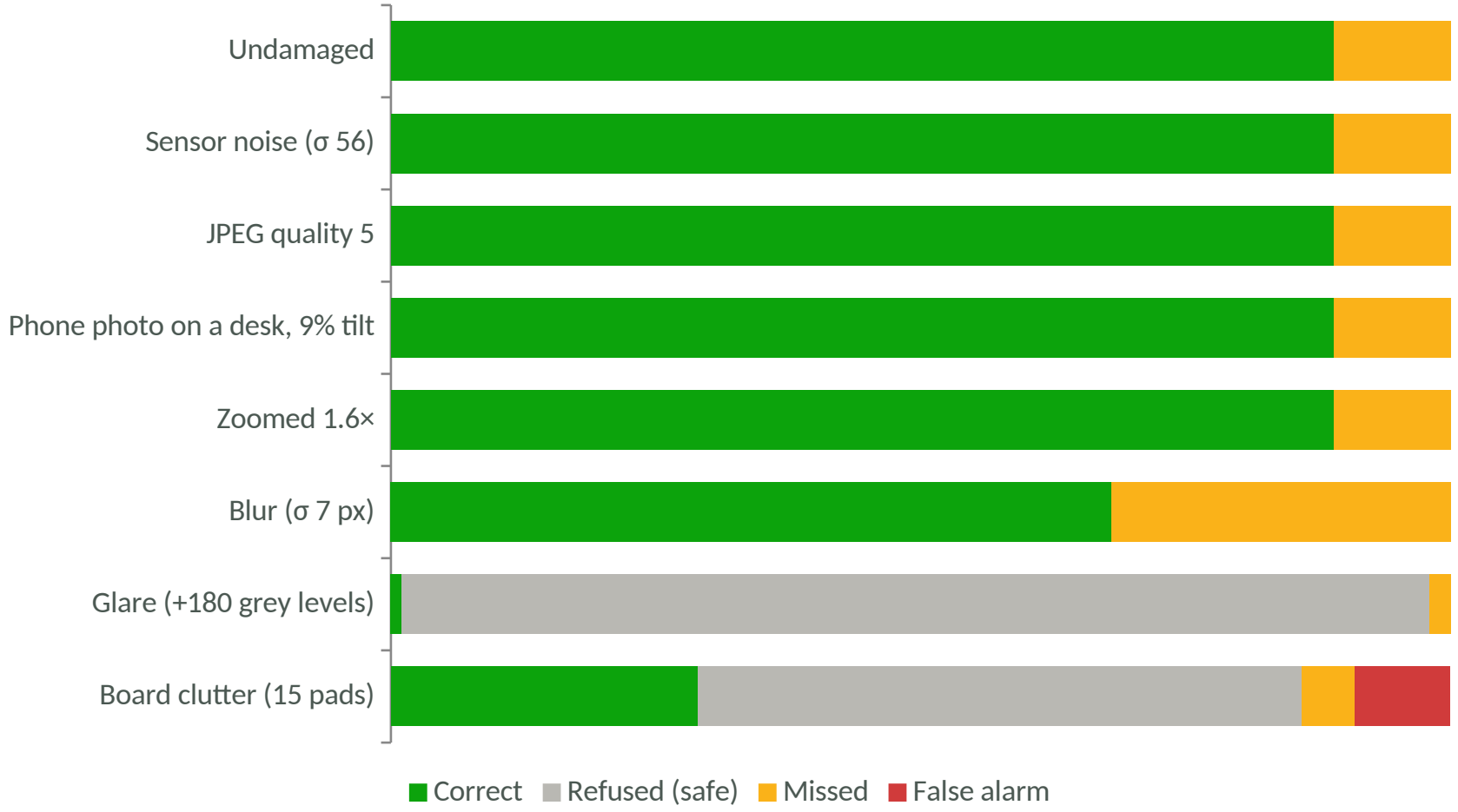
top cause right when half the answers are guesses — with the photo, and without it.

- **With the photo: 81% → 66% → 57%**
- **Questions only: 71% → 54% → 40%**

Chance is 14% (one in seven). 420 simulated cases per point; answers from scenario cards written from the NPL guide, not from the rules.

BACKUP · EVALUATION

It refuses rather than misleads



9%

false alarms under light board clutter — the only tested fault that misleads. The fix is a bare-board reference photo.

Everything else measures correctly or is refused with a reason. 'Missed' is the dispense-order pattern found in about half of carryover photos — the interview then carries it.

BACKUP · INDUSTRY IMPACT

Ready to pilot on an NSW line

Value on day one

- A new technician's first check — ranked, with reasons
- An audit-ready report for every case, printable as PDF
- A troubleshooting history that only keeps proved fixes
- Runs offline on a line PC; a phone camera is enough

Pilot plan

1. Validate on 50–100 real NSW photos per product
2. Subtract a bare-board reference photo to remove clutter false alarms
3. Rules as data: engineers add causes (e.g. up height, NPL §3.3) without code
4. Predict: open a case automatically when a line's size spread starts drifting

BACKUP · WHAT THE NUMBERS MEAN

Score, tier, verdict: three different questions

AFTER THE PHOTO

Quality score

How good are the dots?

20/100

★☆☆☆☆ size, 3 of 15

Decided by: Counts of dots that pass four checks: present, size, shape, position

Detail: B9

AFTER EVERY RANKING

Confidence tier

How far can I trust the ranking?

T1 T2 T3 T4

Decided by: The two brains and two gates, or a check that confirmed the cause

Detail: B4

WHEN THE CASE CLOSES

Verdict

Was the fix proved?

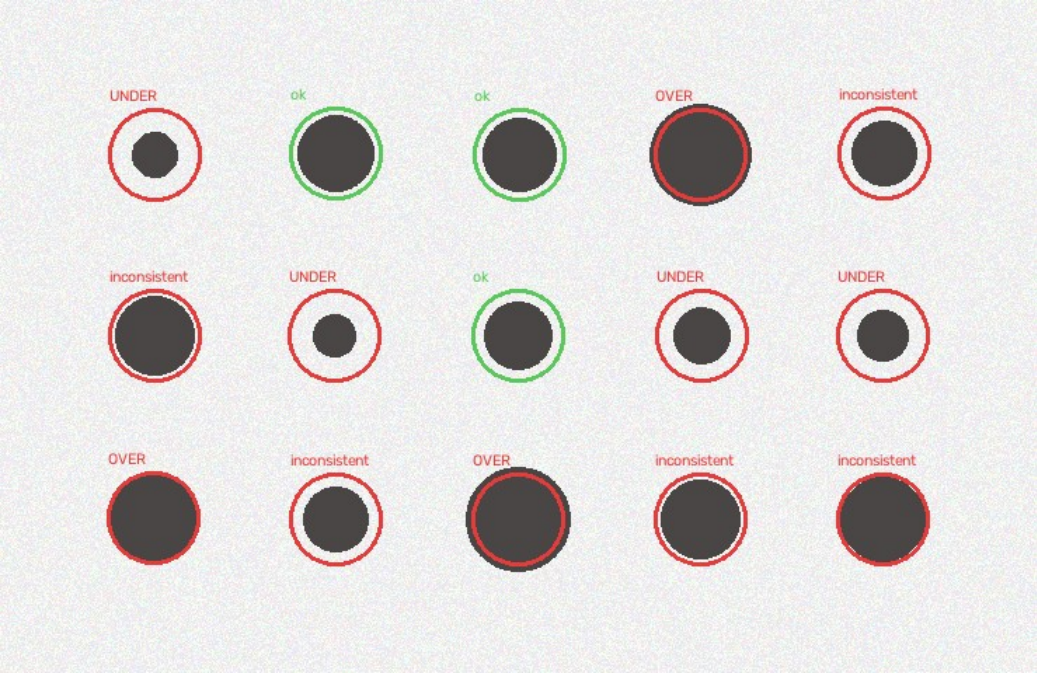
CONFIRMED PLAUSIBLE UNVERIFIED

Decided by: Five conditions, read from the check log and a re-measured photo

Detail: B3

Separate answers: the quality score never sets the tier, and T1 is not CONFIRMED.

A quality score you can check by counting the dots



Real output on a test photo with scattered sizes: the 3 green rings are the only dots within 15% of the nominal size.

Dispensing quality — **20**/100

3 of 15 dots pass all four

Completeness ★★★★★
15 of 15 deposits present

Size consistency ★☆☆☆☆
3 of 15 within 15% of the nominal size

Shape consistency ★★★★★
15 of 15 round

Dispensing position ★★★★★
15 of 15 where the recipe puts them

Stars = share passed × 5, rounded down, so one failure always costs a star

Overall = dots passing all four ÷ all dots: 3 of 15 → 20/100

Rates the dots, not the diagnosis: how far to trust the ranking is the tier's job

Shown when the photo passes the quality check and the job has a recipe; without the expected dot positions there is nothing to rate.